EVM Dashboard — Validation Report

CE 725 Construction Economics and Finance · Course project, Topic 7

Status of this report. All test results below come from actual runs on 21-Oct-2026 (evidence: `reports/test_results.txt`, `reports/benchmark_comparison.md`, `reports/browser_test_log.txt`). The independent benchmark calculations were prepared with AI assistance (Claude) separately from the dashboard code. **The course requires the team to verify every number against a manual benchmark; the Validation Engineer must redo Section 4 on paper and sign Section 15 before submission.** Until then, Section 4 is "prepared, awaiting team verification".

1. Validation objective

To show that every metric the dashboard reports — PV, EV, AC, CV, SV, CPI, SPI, EAC (methods A–C), ETC, VAC, forecast overrun, TCPI to BAC and TCPI to EAC — equals an independently prepared manual result for two reporting periods (course requirement, Topic 7), and that undefined, invalid and edge-case inputs are handled without displaying fabricated numbers.

2. Benchmark project

Synthetic, illustrative G+4 RCC residential block. Four work packages; BAC ₹4,00,00,000 (sum of WP budgets). Two monthly status dates: Period 1 = 31-Jan-2026, Period 2 = 28-Feb-2026. A work-package case was chosen deliberately because it also tests budget weighting, the most common EV error. It also contains cost without progress (WP4, Period 1: ₹1,00,000 spent at 0% complete — a procurement advance), which must not be treated as an error.

3. Input data (cumulative)

Work package	Budget (₹)	P1 planned %	P1 actual %	P1 AC (₹)	P2 planned %	P2 actual %	P2 AC (₹)
WP1 Substructure	80,00,000	100	100	84,00,000	100	100	84,00,000
WP2 RCC superstructure	1,60,00,000	40	30	52,00,000	70	55	98,00,000
WP3 Masonry and finishes	1,20,00,000	0	0	0	10	5	7,00,000
WP4 MEP services	40,00,000	5	0	1,00,000	10	5	3,00,000
Total	**4,00,00,000**						

File: `data/benchmark_case_work_packages.csv`. Assumptions: all values cumulative; actual % is physical progress against budgeted scope; one baseline.

4. Manual calculations (working shown; ₹ lakh = ₹1,00,000)

Period 1 — 31-Jan-2026

- PV = 80×1.00 + 160×0.40 + 120×0 + 40×0.05 = 80 + 64 + 0 + 2 = **146.00 lakh**
- EV = 80×1.00 + 160×0.30 + 120×0 + 40×0 = 80 + 48 = **128.00 lakh**
- AC = 84 + 52 + 0 + 1 = **137.00 lakh**
- CV = 128 − 137 = **−9.00 lakh**; SV = 128 − 146 = **−18.00 lakh**
- CPI = 128/137 = **0.934307**; SPI = 128/146 = **0.876712**
- EAC(A) = BAC/CPI = 400 × 137/128 = **428.125 lakh**; ETC = 428.125 − 137 = 291.125; VAC = −28.125; overrun 28.125/400 = **7.03125%**
- EAC(B) = 137 + (400 − 128)/(CPI×SPI); CPI×SPI = (128×128)/(137×146) = 16,384/20,002 = 0.819118; 272/0.819118 = 272×20,002/16,384 = **332.064453**; EAC(B) = **469.064453 lakh**; ETC 332.064453; VAC −69.064453; overrun **17.266113%**
- EAC(C) = 137 + 272 = **409.00 lakh**; ETC 272; VAC −9; overrun **2.25%**
- TCPI to BAC = (400 − 128)/(400 − 137) = 272/263 = **1.034221**
- TCPI to EAC: A = 272/291.125 = 0.934307 (= CPI ✓); B = 272/332.064453 = 0.819118 (= CPI×SPI ✓); C = 272/272 = 1 ✓

Period 2 — 28-Feb-2026

- PV = 80 + 160×0.70 + 120×0.10 + 40×0.10 = 80 + 112 + 12 + 4 = **208.00 lakh**
- EV = 80 + 160×0.55 + 120×0.05 + 40×0.05 = 80 + 88 + 6 + 2 = **176.00 lakh**
- AC = 84 + 98 + 7 + 3 = **192.00 lakh**
- CV = **−16.00 lakh**; SV = **−32.00 lakh**; CPI = 176/192 = **0.916667**; SPI = 176/208 = **0.846154**
- EAC(A) = 400 × 192/176 = **436.363636 lakh**; ETC 244.363636; VAC −36.363636; overrun **9.090909%**
- EAC(B): CPI×SPI = 30,976/39,936 = 0.775641; 224/0.775641 = 224×39,936/30,976 = **288.793388**; EAC(B) = **480.793388 lakh**; overrun **20.198347%**
- EAC(C) = 192 + 224 = **416.00 lakh**; overrun **4.00%**
- TCPI to BAC = 224/208 = **1.076923**

Weighting check. An unweighted average of Period 2 actual percentages, (100+55+5+5)/4 = 41.25%, would give EV = 165 lakh — ₹11 lakh (6.3%) below the correct budget-weighted 176 lakh. The dashboard must give 176.

Independent cross-checks of the paper working. (a) `benchmark/independent_benchmark_calc.py` recomputes everything with exact fractions and imports no dashboard code (enforced by a test that inspects its import statements); (b) `benchmark/benchmark_workbook.xlsx` computes the same with plain spreadsheet formulas (73 formulas, recalculated in LibreOffice, 0 errors). All three agree to the last digit shown above.

5. Dashboard calculation procedure

The dashboard values were produced through the same path a user follows: read the CSV → map columns → validate → derive PV/EV/AC → calculate (`tests/test_benchmark.py`, `benchmark/compare_with_dashboard.py`). The same project was also entered (i) as pre-calculated PV/EV/AC and (ii) as project-level percentages (36.5%/32% and 52%/44%), to check the three input modes agree. In the browser, the benchmark PV/EV/AC were entered through the manual-entry table and loaded; the Overview showed CPI 0.917, SPI 0.846, EAC(A) ₹436.36 lakh and TCPI 1.077 for Period 2, matching Section 4.

6. Comparison

Tolerances: money ±₹0.01 (one paisa); indices ±1×10⁻⁹; percentages ±1×10⁻⁷. Values were compared at full precision — no rounding before comparison.

- **work_package** input mode: 54 values compared, 54 PASS, 0 FAIL; max |difference| = 7.45e-09
- **precalc** input mode: 54 values compared, 54 PASS, 0 FAIL; max |difference| = 7.45e-09
- **progress** input mode: 54 values compared, 54 PASS, 0 FAIL; max |difference| = 7.45e-09

Full table (work-package input mode; the other two modes give identical results):

Metric	Method	Period	Manual result	Dashboard result	Difference	Tolerance	Status
BAC		2026-01-31	40,000,000.0000	40,000,000.0000	0.00e+00	0.01	PASS
PV		2026-01-31	14,600,000.0000	14,600,000.0000	0.00e+00	0.01	PASS
EV		2026-01-31	12,800,000.0000	12,800,000.0000	0.00e+00	0.01	PASS
AC		2026-01-31	13,700,000.0000	13,700,000.0000	0.00e+00	0.01	PASS
CV		2026-01-31	-900,000.0000	-900,000.0000	0.00e+00	0.01	PASS
SV		2026-01-31	-1,800,000.0000	-1,800,000.0000	0.00e+00	0.01	PASS
CPI		2026-01-31	0.9343065693	0.9343065693	4.31e-11	1e-09	PASS
SPI		2026-01-31	0.8767123288	0.8767123288	-3.29e-11	1e-09	PASS
TCPI_BAC		2026-01-31	1.0342205323	1.0342205323	1.94e-11	1e-09	PASS
EAC	A	2026-01-31	42,812,500.0000	42,812,500.0000	0.00e+00	0.01	PASS
ETC	A	2026-01-31	29,112,500.0000	29,112,500.0000	0.00e+00	0.01	PASS
VAC	A	2026-01-31	-2,812,500.0000	-2,812,500.0000	0.00e+00	0.01	PASS
OVERRUN	A	2026-01-31	2,812,500.0000	2,812,500.0000	0.00e+00	0.01	PASS
OVERRUN_PCT	A	2026-01-31	7.03125000	7.03125000	0.00e+00	1e-07	PASS
TCPI_EAC	A	2026-01-31	0.9343065693	0.9343065693	4.31e-11	1e-09	PASS
EAC	B	2026-01-31	46,906,445.3125	46,906,445.3125	0.00e+00	0.01	PASS
ETC	B	2026-01-31	33,206,445.3125	33,206,445.3125	0.00e+00	0.01	PASS
VAC	B	2026-01-31	-6,906,445.3125	-6,906,445.3125	0.00e+00	0.01	PASS
OVERRUN	B	2026-01-31	6,906,445.3125	6,906,445.3125	0.00e+00	0.01	PASS

Metric	Method	Period	Manual result	Dashboard result	Difference	Tolerance	Status
OVERRUN_PCT	B	2026-01-31	17.26611328	17.26611328	5.00e-11	1e-07	PASS
TCPI_EAC	B	2026-01-31	0.8191180882	0.8191180882	-8.82e-12	1e-09	PASS
EAC	C	2026-01-31	40,900,000.0000	40,900,000.0000	0.00e+00	0.01	PASS
ETC	C	2026-01-31	27,200,000.0000	27,200,000.0000	0.00e+00	0.01	PASS
VAC	C	2026-01-31	-900,000.0000	-900,000.0000	0.00e+00	0.01	PASS
OVERRUN	C	2026-01-31	900,000.0000	900,000.0000	0.00e+00	0.01	PASS
OVERRUN_PCT	C	2026-01-31	2.25000000	2.25000000	0.00e+00	1e-07	PASS
TCPI_EAC	C	2026-01-31	1.0000000000	1.0000000000	0.00e+00	1e-09	PASS
BAC		2026-02-28	40,000,000.0000	40,000,000.0000	0.00e+00	0.01	PASS
PV		2026-02-28	20,800,000.0000	20,800,000.0000	0.00e+00	0.01	PASS
EV		2026-02-28	17,600,000.0000	17,600,000.0000	0.00e+00	0.01	PASS
AC		2026-02-28	19,200,000.0000	19,200,000.0000	0.00e+00	0.01	PASS
CV		2026-02-28	-1,600,000.0000	-1,600,000.0000	0.00e+00	0.01	PASS
SV		2026-02-28	-3,200,000.0000	-3,200,000.0000	0.00e+00	0.01	PASS
CPI		2026-02-28	0.9166666667	0.9166666667	-3.33e-11	1e-09	PASS
SPI		2026-02-28	0.8461538462	0.8461538462	-4.62e-11	1e-09	PASS
TCPI_BAC		2026-02-28	1.0769230769	1.0769230769	2.31e-11	1e-09	PASS
EAC	A	2026-02-28	43,636,363.6364	43,636,363.6364	7.45e-09	0.01	PASS
ETC	A	2026-02-28	24,436,363.6364	24,436,363.6364	0.00e+00	0.01	PASS
VAC	A	2026-02-28	-3,636,363.6364	-3,636,363.6364	-4.66e-09	0.01	PASS
OVERRUN	A	2026-02-28	3,636,363.6364	3,636,363.6364	4.66e-09	0.01	PASS
OVERRUN_PCT	A	2026-02-28	9.09090909	9.09090909	9.10e-12	1e-07	PASS
TCPI_EAC	A	2026-02-28	0.9166666667	0.9166666667	-3.33e-11	1e-09	PASS
EAC	B	2026-02-	48,079,338.84	48,079,338.84	7.45e-09	0.01	PASS

Metric	Method	Period	Manual result	Dashboard result	Difference	Tolerance	Status
		28	30	30			
ETC	B	2026-02-28	28,879,338.8430	28,879,338.8430	7.45e-09	0.01	PASS
VAC	B	2026-02-28	-8,079,338.8430	-8,079,338.8430	-7.45e-09	0.01	PASS
OVERRUN	B	2026-02-28	8,079,338.8430	8,079,338.8430	7.45e-09	0.01	PASS
OVERRUN_PCT	B	2026-02-28	20.19834711	20.19834711	3.80e-11	1e-07	PASS
TCPI_EAC	B	2026-02-28	0.7756410256	0.7756410256	4.10e-11	1e-09	PASS
EAC	C	2026-02-28	41,600,000.0000	41,600,000.0000	0.00e+00	0.01	PASS
ETC	C	2026-02-28	22,400,000.0000	22,400,000.0000	0.00e+00	0.01	PASS
VAC	C	2026-02-28	-1,600,000.0000	-1,600,000.0000	0.00e+00	0.01	PASS
OVERRUN	C	2026-02-28	1,600,000.0000	1,600,000.0000	0.00e+00	0.01	PASS
OVERRUN_PCT	C	2026-02-28	4.00000000	4.00000000	0.00e+00	1e-07	PASS
TCPI_EAC	C	2026-02-28	1.0000000000	1.0000000000	0.00e+00	1e-09	PASS

7. Discrepancies and defects found

No numerical discrepancy was found between the manual benchmark and the dashboard (maximum |difference| = ₹7.5×10⁻⁹, floating-point representation). Because a first-run match does not prove the tests can detect errors, mutation testing was performed (Section 12). The following genuine defects and findings arose during development and testing:

#	Found by	Defect / finding	Root cause	Correction	Retest
D1	Unit test run	Test “expected file independent of src” failed although the benchmark script imports nothing from the app	Test searched the text for “from src” and matched a phrase in the script’s docstring (defect in the test, not the app)	Test rewritten to inspect actual import statements (AST)	Pass
D2	Browser test	After a rejected upload, the next (valid) file loaded under the project name typed for the rejected file, not the name in the new file	Settings widgets kept their state across files (same widget keys)	Widgets keyed to the specific file and mode	Pass: second upload took the file’s own name
D3	Browser test (screenshot review)	A non-numeric AC produced two errors for one cell (“not a number”	Missing-value check re-flagged a cell already rejected by	Missing check skips cells already in error	Pass: 5 errors instead of 6

#	Found by	Defect / finding	Root cause	Correction	Retest
		and “missing”)	parsing		
D4	Screenshot review	A warning named the internal field `planned_pct` instead of the user’s column `planned_progress_pct`	Internal column name passed as field label	Mapped to user-facing names	Pass
D5	Data generation	Progress-based demo file EV for 31-Aug-2026 = ₹7,02,50,000.40 vs ₹7,02,50,000 from WP data	Percentages stored to 6 decimals	Not a code defect; documented as limitation (rounded % inputs lose precision)	Recorded
D6	Screenshot review	Signboard clipped, KPI labels truncated, BAC/EAC chart labels overlapping, CPI/SPI axis cutting off the 1.00 line, preview dates showing “00:00:00”	Layout defaults	Labels shortened, annotations repositioned, axis range includes 1.00 and the critical line, dates formatted	Pass (screenshots in `reports/screenshots/`)

8–10. Root causes, corrections and retest

Given per defect in Section 7. After all corrections the complete suite was rerun: **79 passed, 0 failed** (`reports/test_results.txt`), and the browser run was repeated (`reports/browser_test_log.txt`).

11. Edge-case and workflow testing

Case	Expected behaviour	Evidence	Result
AC = 0	CPI undefined (not 0), reason shown; CV defined	`test_cpi_undefined_when_ac_zero`	Pass
PV = 0	SPI undefined	`test_spi_undefined_when_pv_zero`	Pass
EV = 0	CPI = 0; EAC A and B undefined; C defined	`test_zero_earned_value_…`	Pass
AC = BAC with work remaining	TCPI to BAC undefined (“no longer possible”)	`test_tcpi_bac_undefined_when_ac_equals_bac`	Pass
AC > BAC	TCPI to BAC undefined, never negative	`test_tcpi_bac_undefined_when_budget_exceeded`	Pass
EAC below / above BAC	VAC positive / negative	`test_eac_below_bac…`, `test_eac_above_bac`	Pass
Project complete (EV = PV = BAC)	SPI = 1, ETC = 0, TCPI N/A	`test_project_completion`	Pass
Missing reporting period	Warning; chart line broken	`test_missing_monthly_period…`;	Pass

Case	Expected behaviour	Evidence	Result
		`with_gap_breaks`	
Negative cumulative AC / BAC	Blocking error	`test_negative_cumulative_ac_is_error`, `test_missing_project_name_and_bad_bac`	Pass
Progress < 0 or > 100	Blocking error	`test_progress_out_of_range`	Pass
Decreasing cumulative values	Warning, value unchanged	`test_decreasing_cumulative_progress…`	Pass
Cost with no new progress	No error	`test_cost_without_progress_is_not_an_error`	Pass
Cumulative and period AC inconsistent	Warning; cumulative used, never summed	`test_cumulative_and_period_ac_inconsistent`	Pass
Missing value	Error, never zero	`test_missing_value_is_error_not_zero`	Pass
Unweighted WP average	Budget-weighted EV used	`test_work_package_ev_is_budget_weighted…`	Pass
All 4 forecast methods, all 9 dates, all units	Render without error; method D “N/A” until ETC entered	`tests/test_app.py`	Pass
Invalid file upload (browser)	Errors listed by row; no load button; current project unchanged	browser log	Pass
Valid CSV upload (browser)	Loads after confirmation; CPI 0.947	browser log	Pass
Manual entry → load (browser)	Benchmark P2 values reproduced	browser log	Pass
Editing a single cell in the manual table (browser)	Edit re-validates	Scripted edit did not register in the canvas grid	**Not verified** — team to check by hand (2 min): change an AC to −5 and confirm a blocking error appears
Upload ↔ manual entry equivalence	Identical results	`test_manual_entry_and_upload_give_identical_results`	Pass
Older library versions	Suite passes	pandas 2.2.3 / Streamlit 1.50 / Plotly 5.24 venv: 79 passed	Pass

12. Mutation testing (can the tests catch typical errors?)

Each error below was injected into a temporary copy of the code; the suite was run; the copy discarded.

Injected error	Tests failing
SPI inverted (PV/EV)	8
CV sign reversed (AC − EV)	6
EAC(B) written as BAC/(CPI×SPI)	5
TCPI to BAC forced to 1	4
EV from an unweighted average of WP percentages	2
Period AC used as if cumulative	1

Injected error	Tests failing
CPI returned as 0 when AC = 0	2

All seven were detected.

13. Remaining limitations of the validation

- One synthetic benchmark (two periods) — not a real project; practitioner review is still recommended.
- Method D (independent ETC) is checked by unit and interface tests, not by the paper benchmark (it is a user input, not a formula).
- Charts and commentary were checked by inspection and interface tests, not against an independent reference.
- Manual-table cell editing in the browser not verified by automation (Section 11).

14. Conclusion

For the benchmark case, all 54 benchmark values per input mode (162 comparisons) match the independently prepared results within one paisa / 10^{-9}, with no discrepancies; undefined and edge cases are reported as undefined or flagged rather than shown as numbers; and the test suite detects all seven typical formula errors injected. Subject to the team's own paper verification (Section 15), the calculation engine is considered validated for the scope tested.

15. Team verification sign-off

Check	Done by	Date	Signature
Period 1 recalculated on paper without the tool; matches Section 4			
Period 2 recalculated on paper without the tool; matches Section 4			
`python -m pytest` run on a team laptop: 79 passed			
Manual-table cell edit check (Section 11) done			
Any discrepancy found by the team recorded here and in the AI usage log			